



Gennoor Tech
TRAIN. INNOVATE. BUILD.

5-DAY HANDS-ON BOOTCAMP · 40 HOURS

AI Development Framework

Build AI solutions with LangChain, Semantic Kernel, and more

5 40 15+ 5+

DAYS HOURS HANDS-ON LABS PROJECTS

www.gennoor.com · enquiry@gennoor.com

Course Overview

This 5-day hands-on bootcamp covers everything from foundational concepts to advanced techniques, combining lectures, live demos, hands-on labs, and collaborative workshops for maximum learning impact.

Prerequisites: Python (intermediate), REST APIs, basic Git knowledge.

Who Is This For?



Learning Outcomes

- 1 **Integrate LLM APIs** — Azure OpenAI, Claude, Gemini, Ollama
- 2 **Master orchestration frameworks** — LangChain, Semantic Kernel, CrewAI
- 3 **Implement production RAG** — with vector databases and hybrid search
- 4 **Build full-stack AI** — apps with FastAPI, Streamlit, and Gradio
- 5 **Design AI agents**, — MCP servers, and workflow automation

5-Day Curriculum

Interactive format with live demos, hands-on labs, and collaborative workshops. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

LLM APIs & Prompt Engineering for Developers

HANDS-ON LABS + WORKSHOP

- SESSION A
- LLM APIs
- Azure OpenAI, Claude API, Gemini API — unified patterns
 - Streaming, function calling, structured output (JSON mode)
 - Open-source models via Ollama (Llama, Mistral, Qwen)

- SESSION B
- Prompt Engineering for Developers
- Token management, cost tracking, rate limit handling
 - Lab: Multi-provider LLM gateway with fallback logic

02

LangChain, Semantic Kernel & Orchestration

DEMOS + LABS + DISCUSSION

- SESSION A
- LangChain, Semantic Kernel
- LangChain: chains, prompts, tools, memory
 - LangGraph: stateful graph workflows, conditional routing
 - Semantic Kernel: plugins, planners, .NET and Python SDKs

- SESSION B
- Orchestration
- CrewAI and AutoGen: multi-agent patterns
 - Lab: Same app in LangChain vs Semantic Kernel — comparison

03

RAG Pipelines & Vector Databases

LECTURE + HANDS-ON LABS

- SESSION A
- RAG Pipelines
- Chunking strategies: fixed, recursive, semantic, document-aware
 - Embedding models: OpenAI, Cohere, sentence-transformers
 - FAISS, Chroma, Pinecone, Azure AI Search — comparison

- SESSION B
- Vector Databases
- Hybrid search, semantic ranking, reranking
 - Lab: Enterprise document Q&A with citation tracking

04

Full-Stack AI Apps

HANDS-ON LABS + WORKSHOP

- SESSION A
- Full-Stack AI Apps
- FastAPI: async endpoints, WebSocket streaming, authentication
 - MCP Server development: tools, resources, prompts protocol
 - Streamlit: rapid AI app prototyping with chat interfaces

- SESSION B
- Hands-On & Practice
- Gradio: model demos, file upload, multimodal interfaces
 - Lab: Document Q&A app (Streamlit + RAG + FastAPI)

05

Agents, Automation & Capstone

DEMOS + LABS + DISCUSSION

- SESSION A
- Agents, Automation
- Tool-calling agents with function calling and MCP
 - Workflow automation: N8N, Power Automate, Copilot Studio
 - Docker containerization and cloud deployment

- SESSION B
- Capstone
- Monitoring, logging, and cost optimization
 - Capstone: Full-stack AI application

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools demonstrated and practiced hands-on during the bootcamp

AutoGen

Azure AI Search

Azure AI Services

Azure OpenAI

ChromaDB

Claude (Anthropic)

CrewAI

Docker

FAISS

FastAPI

Google Gemini

LangChain

LangGraph

Microsoft Copilot

Microsoft Copilot Studio

N8N

Ollama

Power Automate

Python

Semantic Kernel

Streamlit

What You Take Home

 **6 AI Applications (complete source code)**
Included with course completion

 **Framework Decision Matrix (LangChain vs SK vs CrewAI)**
Included with course completion

 **AI App Starter Templates (FastAPI + Streamlit + Docker)**
Included with course completion

 **Architecture Patterns Guide (RAG, agents, caching, streaming)**
Included with course completion

Certification of Completion

**Gennoor Tech — Certificate of Completion**
Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "**AI Development Framework**". Digitally signed by the Founder & Director. Validates proficiency in all course topics and hands-on competency.

Recommended Next Certifications

AI-103
Azure AI Solutions

AZ-204
Azure Developer Associate

AI-901
Azure AI Fundamentals (New 2026)

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited
Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of **16+ active Microsoft certifications** experience. Delivered AI and technology programs for Microsoft, EY, IBM, Boeing, including AB-100 (Agentic AI), AI-103, AZ-400, Saudi Aramco, HDFC Bank, and 80+ organizations across 6+ countries including the AI-300, PL-400, and GitHub certifications. 10-day AI Leadership program for MCIT Saudi Arabia.
Builder of open-source MCP servers and Copilot Studio integrations. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options ■ **On-Site** — customized at your office | ■ **Virtual Live** — interactive via Teams/Zoom | ■ **International** — GCC, Africa, APAC, UK



Ready to Start Your AI Development Framework Journey?

Contact us to discuss a customized program
for your organization or register for the next open batch.

🌐 www.gennoor.com

✉ enquiry@gennoor.com

☎ +91 9326 352 241

🔗 linkedin.com/company/gennoor-tech-private-limited

▶ YouTube — @GennoorTech

Gennoor Tech

TRAIN. INNOVATE. BUILD.